

Heisenberg Group and Reality

P. Deligne

1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ) , the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$\tau : (x, y, \lambda) \mapsto (-x, -y, \lambda) \quad (1.2)$$

is an automorphism of H .

The coordinate system $x, y, \lambda : H \rightarrow \mathbb{C}^2 \times \mathbb{C}^*$ exhibits the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section, $\tau(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, by means of this section, a lift to a one-parameter subgroup of H , again denoted by D . The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$, with

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H , and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which as a set is identified with $\mathbb{R}^2 \times U^1$.

It is known, by the Stone-von Neumann theorem [2], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V for which $\lambda \in U^1$ acts by multiplication by λ . In the Schrödinger model of V , one has $V = L^2(\mathbb{R})$, and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (x, 0, 1) : f(z) &\mapsto f(z + x), \\ (0, y, 1) : f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of V one deduces a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V , compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [3], D. Shale [1]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathcal{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V , are the L^2 functions on $\mathbb{R}$ that extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\mathrm{Im} z| < A_1$, one has, for all $A_2 \in \mathbb{R}$,

$$|f(z)| \leq O(\exp(-A_2|z|)).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol} , one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V . If one did, the projective action of $\mathrm{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\mathrm{SL}(2, \mathbb{C})$ on V^{hol} ; since $\mathrm{SL}(2, \mathbb{C})$ is simply connected, this action would lift to an honest action of $\mathrm{SL}(2, \mathbb{C})$, and the projective action of $\mathrm{SL}(2, \mathbb{R})$ would lift to an honest action. This is not the case: only the double cover of $\mathrm{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

2

Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines of $\mathbb{C}^2$. For $D \in S$, lifted in H as in no. 1, let W_D be the space of holomorphic functions φ on H satisfying

$$\varphi(\lambda dh) = \lambda \varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by $g : \varphi(h) \mapsto \varphi(hg)$. For $E \neq D$, restriction to E is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \quad (2.1)$$

where $\mathcal{O}(E)$ denotes the space of holomorphic functions on E .

If R is a holomorphic representation of H , on which each $\lambda \in \mathbb{C}^*$ acts by λ , and if ω is a D -invariant linear form on R , then

$$w \mapsto \omega(hw) \quad (2.2)$$

is a morphism from R to W_D .

3. Example

Let V , as in nos. 1 and 2, be the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathcal{S} \subset V$ be the space of its C^∞ vectors, and let $\mathcal{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\text{Im } t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t : f \mapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t) dx$$

is defined on V , and even on $\mathcal{S}'$. The form ω_t is a holomorphic vector of the dual of V , fixed by D_t . In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t , the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2), ω_t defines

$$\Omega_t : V^{\text{hol}} \longrightarrow W_{D_t},$$

which extends to V and even to $\mathcal{S}'$. Let E be the x -axis. The map $r_E \Omega_t : V \rightarrow \mathcal{O}(E)$ sends $f \in V$ to

$$f(x, t) = r_E \Omega_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\left(\partial_t - \frac{1}{4\pi i} \partial_x^2 \right) f(x, t) = 0.$$

For real t , the form ω_t remains defined on $\mathcal{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $f \mapsto \int f(x) dx$. The homomorphism Ω_t remains defined on V^{hol} . For $t = 0$, $r_E \Omega_0$ is, in the Schrödinger model, the identity.

Assume $\text{Im } t > 0$. Let $\overline{D}_t$ be the complex conjugate of D_t . In the affine line $S - \{\overline{D}_t\}$, the “half-plane” $\{D_u \mid \text{Im } u > 0\}$ is a disk with center D_t . The map $r_{\overline{D}_t} \circ \Omega_t : V \rightarrow W_{D_t} \rightarrow \mathcal{O}(\overline{D}_t)$ gives the holomorphic model of V : the action of H is given by

$$\begin{aligned} d \in D_t : \quad & f(x) \mapsto f(x + d), \\ e \in \overline{D}_t : \quad & f(x) \mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned}$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d)\overline{f(\bar{d})}\exp(2\pi i\langle d, \bar{d} \rangle)$$

is invariant under $H_{\mathbb{R}}$; moreover $r_{\bar{D}_t} \circ \Omega_t$ identifies V with the space of f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i\langle d, \bar{d} \rangle)$, the commutator of d and $\bar{d}$, is

$$\exp(2\pi i\langle d, \bar{d} \rangle) = \exp(2\pi i(t - \bar{t})|z|^2).$$

4

Fix E . The isomorphisms r_E of (2.1) make it possible to regard the W_D for $D \neq E$ as a family of representations of H operating in the fixed space $\mathcal{O}(E)$, with the action of H depending holomorphically on D . Let us make explicit the dependence on E of r_E . Choose coordinates as in no. 1, with E the x -axis. Let E' be generated by $(1, u)$ and D by $(-t, 1)$. We use x as coordinate on both E and E' . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi i u x x')$. Hence $r_{E'} r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$, with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'}\left(\frac{x}{1+tu}\right) = \exp\left(\frac{\pi i u}{1+tu}x^2\right) f_E(x). \quad (4.1)$$

This formula is holomorphic in t : the W_D form a holomorphic vector bundle $\mathcal{W}$ of infinite dimension over S , on which H acts.

To understand (4.1), one may observe that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \neq D$ and of the coordinate z on E :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O(\exp(A|z|^2)). \quad (4.2)$$

Let $W_D^0 \subset W_D$ be defined by this condition. The W_D^0 form a holomorphic sub-bundle $\mathcal{W}^0$ of the bundle $\mathcal{W}$ in holomorphic representations of H .

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f :

- (i)_T There exists a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$, with initial condition $g(z, 0) = f(z)$.
- (ii)_T $|f(z)| \leq O(\exp(|z|^2/4T))$.
- (iii)_T $\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|dz \wedge d\bar{z}| < \infty$.

If $T < T_1$, each of $(i)_{T_1}$, $(ii)_{T_1}$, $(iii)_{T_1}$ implies each of $(i)_T$, $(ii)_T$, $(iii)_T$.

Proof. $(i)_{T_1} \Rightarrow (ii)_T$. The heat equation remains satisfied by the derivatives of g , and therefore

$$\partial_z^{2n} g = \partial_t^n g, \quad \partial_z^{2n+1} g = \partial_t^n (\partial_z g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \quad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for $\partial_z^{2n+1}g$. The Taylor series of f then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!} \frac{1}{T_2^n} |z|^{2n}\right).$$

One has

$$\frac{1}{2n+1} 2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n} \frac{1}{T^n} \frac{|z|^{2n}}{n!}\right),$$

and this sum is $\exp(|z|^2/4T)$.

$(ii)_{T_1} \Rightarrow (i)_T$. It suffices to take g given by the kernel of the heat equation:

$$g(z, t) = \int_{\sqrt{t}\mathbb{R}} f(z+x) \frac{1}{2\sqrt{\pi t}} \exp(-x^2/4t) dx.$$

Changing $\sqrt{t}$ to $-\sqrt{t}$ changes the orientation of the cycle $\sqrt{t}\mathbb{R}$ and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: $(ii)_{T_1} \Rightarrow (iii)_T$, and $(iii)_{T_1} \Rightarrow (ii)_T$ is verified by writing $f(z)$ as the mean of its values on a disk with center z and radius $1/|z|$, and estimating that mean by Cauchy–Schwarz.

6

In homage to the Stone-von Neumann theorem, the bundle $\mathcal{W}$ over S is, projectively, equipped with a holomorphic connection for which the action of H is horizontal.

Let us specify “projectively.” On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L endowed with an isomorphism $L^{\otimes 2} \simeq \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity ± 1 . This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathcal{W}$, but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \quad (6.1)$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and if ∇ is a connection on $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$, then its transport to $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$ of square roots of $\mathcal{O}(-1)$ does not depend on the choice of a , which can be changed locally only by a constant ± 1 . Thus the notion of a “connection on $\mathcal{W}(-\frac{1}{2})$ ” is well defined independently of the global existence, or the choice, of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D . For a local section s of $\mathcal{W}(-\frac{1}{2})$, the problem is to define $\nabla_t s$ in the fiber of $\mathcal{W}(-\frac{1}{2})$ at D . We proceed as follows: after making an auxiliary choice u , one defines $\nabla_t^u s$ for s a local section of $\mathcal{W}$ or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u , with dependence in u of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where a takes opposite values for $\mathcal{W}$ and for $\mathcal{O}(-\frac{1}{2})$. Hence, for $s = s_1 s_2$, with s_1 a local section of $\mathcal{W}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u .

Let $D \in S$ be a line of $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t : D \rightarrow \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D)^{\otimes 2}$: to $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$ one associates $t(d) = \langle u, d \rangle v'$, where u is any representative of u' . The auxiliary datum will be a lift $\tilde{t} : D \rightarrow \mathbb{C}^2$ of t , or equivalently a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it $\tilde{t} = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let δ_z be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of H . For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y \rangle \delta_{(0,0,1)}. \quad (6.2)$$

Let $\mathcal{U}_1$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta_{(0,0,1)} = 1$. In $\mathcal{U}_1$, (6.2) implies that, for $d \in D$,

$$\left[-\frac{\delta_u^2}{4\pi i}, \delta_d \right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d \rangle \delta_u = \langle u, d \rangle \delta_u = \delta_{t(d)}. \quad (6.3)$$

Let us compute to first order around D . For $\varepsilon^2 = 0$ and $\varphi(g)$ in W_D , so $\delta_d \varphi = 0$, one has

$$\delta_{d+\varepsilon t(d)} \left[\varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = \varepsilon \left[\delta_{t(d)} \varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = 0,$$

because $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$, by (6.3). To first order around D , the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle $\mathcal{W}$. If $\varphi(g, D')$ is a local section of $\mathcal{W}$, this permits one to define the derivative $\nabla_t^u \varphi$ in $\mathcal{W}_D$ by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on u . If u is replaced by $u + x$ with $x \in D$, then in $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u \delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u \delta_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For $\varphi \in W_D$, $\delta_x \varphi = 0$, and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

whence

$$\frac{1}{4\pi i} \delta_{u+x}^2 \varphi = \frac{1}{4\pi i} \delta_u^2 \varphi + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of u also makes it possible to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d \rangle u$ to obtain a homogeneous function of degree 1 on D . When u is changed into $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2} \nabla_t^u(s) s^{-3/2},$$

and, for the dependence on u , s now being a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x \rangle s. \quad (6.9)$$

The dependences on u in (6.8) and (6.9) cancel to provide a derivation ∇_t for $\mathcal{W}(-\frac{1}{2})$. Since the connection ∇ is defined in terms of right-invariant vector fields on H , it commutes with the action of H .

7

Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let E be the x -axis. On S , $\mathcal{O}(1)$ has a section s with a simple zero at E . It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1} . By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{W}$. We calculate it.

Let D_t be the line generated by $(-t, 1)$. By r_E of (2.1), a local section of $\mathcal{W}$ is identified with a function $f(z, t)$ holomorphic in z and t . We compute its derivative with respect to t . In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1, 0), d \rangle (1, 0).$$

The corresponding derivation of $\mathcal{O}(1)$ annihilates s , so that simply

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through $s \in W_D^0(-\frac{1}{2})$ there passes a horizontal local section of $\mathcal{W}(-\frac{1}{2})$. It is automatically a section of $\mathcal{W}^0(-\frac{1}{2})$. It exists in a neighborhood of D that is larger the slower the growth of s . Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection ∇ on $\mathcal{W}^0(-\frac{1}{2})$ defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\frac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of $\mathcal{W}^0(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{W}^0)$ on S has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{W}^0)$.

First proof. Divide S into two hemispheres S_1 and S_2 , and choose $\mathcal{O}(-\frac{1}{2})_i$ on S_i . On $S_1 \cap S_2$ one has locally an isomorphism $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$, defined up to sign, but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$: the monodromy is -1 . If there existed global horizontal projective sections of $\mathbb{P}(\mathcal{W}^0)$, there would exist horizontal sections f_1, f_2 of $\mathcal{W}^0(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would coincide up to sign. On $S_1 \cap S_2$

this would make it possible to normalize $\alpha : \mathcal{O}(-\frac{1}{2})_1 \rightarrow \mathcal{O}(-\frac{1}{2})_2$ by $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x -axis and D the y -axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D . By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{W}^0)$ passing through f , it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon|z|^2)). \quad (8.1)$$

If E' is another line, then by (4.1) the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon|z|^2)) \quad (8.2)$$

with suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$; hence in all directions except those for which az^2 is purely imaginary, g decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf, g decreases exponentially quadratically in all directions, hence is zero, and therefore $f = 0$.

9

In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. The commutator of d and $\bar{d}$ is

$$\langle d, \bar{d} \rangle = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \bar{d} \rangle > 1 \iff \operatorname{Im}(t) > 0.$$

The lines D such that, for d generating D , one has $\langle d, \bar{d} \rangle > 1$, therefore form a disk in S , called the disk attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ for which $\langle \cdot, \cdot \rangle$ is real on L form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, with $\operatorname{SL}(2, \mathbb{R})$ as the stabilizer of $\mathbb{R}^2$. Each defines a real structure $H_L = L \cdot U^1$ on H , hence a complex conjugation σ_L with fixed points LU^1 .

The disks in S likewise form a homogeneous space under $\operatorname{SL}(2, \mathbb{C})$, the stabilizer of the one associated to $H_{\mathbb{R}}$ being $\operatorname{SL}(2, \mathbb{R})$. The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S .

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

10

The calculations of no. 3 admit the following interpretation. Let X be an open disk of S , corresponding to a real structure L , whence real forms H_L of H and $\operatorname{SL}(L)$ of $\operatorname{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ has an H_L -invariant pre-Hilbert structure; its completion $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$ is the irreducible unitary representation of H_L . One has

$$\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\frac{1}{2})^{\nabla=0}).$$

The double cover $\mathrm{SL}(L)^\sim$ of $\mathrm{SL}(L)$ acts on $\mathcal{O}(-\frac{1}{2})|_X$ compatibly with its action on $\mathcal{O}(-1)$; hence $\mathrm{SL}(L)^\sim$ acts on $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$. This is the action that normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)$ on H_L .

The subrepresentation $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ may be interpreted as the space of real analytic vectors for the action of $\mathrm{SL}(L)^\sim$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

11

Everything preceding generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle \cdot, \cdot \rangle$ real on $\mathbb{R}^{2n}$, replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle)),$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \bar{d} \rangle| > 1$ for $d \neq 0$ in D . The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ must be replaced by $\det(D)^{1/2}$.

12. Remark

Beside representations of H in which $\mathbb{C}^*$ acts by multiplication by λ , consider likewise representations in which the action is by λ^{-1} . One defines, as in no. 2, a holomorphic bundle $\mathcal{W}_-^0$ of such representations over S , and on $\mathcal{W}_-^0(-\frac{1}{2})$ one again has a connection, given by a heat equation.

On a connected simply connected open subset U of S , consider $W = \mathcal{W}^0(-\frac{1}{2})$, its connection, and the action of H . In $\Gamma(U, W)^{\nabla=0}$, for each $x \in U$ there is a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(e)$ for $f \in \mathcal{W}_x^0$. For each $x \notin U$, there exists a vector $v(x)$ fixed by D_x ; in the Schrödinger model, and for x corresponding to $E =$ the x -axis, it is the constant 1. It is unique up to a factor.

Fix the real structure $H_{\mathbb{R}}$, whence a real structure $S_{\mathbb{R}} \subset S$ (a circle), and a decomposition of $S - S_{\mathbb{R}}$ into two disks U_1, U_2 . Then:

- (a) The complex conjugate of $\Gamma(U_1, W)^{\nabla=0}$ is $\Gamma(U_2, W_-)^{\nabla=0}$.
- (b) The Hermitian form on $\Gamma(U_1, W)^{\nabla=0}$, invariant under $H_{\mathbb{R}}$, is reinterpreted as a bilinear pairing between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, invariant under H .

Question. Let C be a Jordan curve in S , separating $S - C$ into two open sets U_1 and U_2 . Does there exist a unique separating H -invariant bilinear form pairing $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$, induced by a perfect duality between $\Gamma(U_1, W)^{\nabla=0}$ and $\Gamma(U_2, W_-)^{\nabla=0}$?

References

- [1] D. Shale, *Linear symmetries of free Boson fields*, Trans. Amer. Math. Soc. 103 (1962), 149–167.
- [2] J. von Neumann, *Die Eindeutigkeit der Schrödingerschen operatoren*, Ann. Mat. Pura Appl. (4) 104 (1931), 570–578.
- [3] A. Weil, *Sur certains groupes d'opérateurs unitaires*, Acta Math. 111 (1964), 143–211.

Real Mixed Hodge Structures

Pierre Deligne

English working translation; formulas, filtrations, and numbering retained.

The real mixed Hodge structures of §2 form a rigid abelian tensor category, and the functor “underlying real vector space” is a fiber functor. The functor Gr^W is another one. We compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposite filtrations

1.1

This paragraph is a variation on §1.2 of P. Deligne, *Théorie de Hodge II*, Publ. Math. IHES 40, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \bar{F}$. Here “filtration” means finite filtration; W is increasing, and F and $\bar{F}$ are decreasing. The condition of opposition is

$$\mathrm{Gr}_n^W \mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q A = 0 \quad \text{for } n \neq p + q.$$

For a bigraded object $B = \bigoplus B^{p,q}$, we write $W, F_W, \bar{F}_W$ for the three opposite filtrations

$$W_n B = \bigoplus_{p+q \leq n} B^{p,q}, \quad F_W^i B = \bigoplus_{p \geq i} B^{p,q}, \quad \bar{F}_W^i B = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, saying that $W, F, \bar{F}$ are opposite on A is equivalent to saying that, for each n , $\mathrm{Gr}_n^W A$ has a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations induced by F and $\bar{F}$ on $\mathrm{Gr}_n^W A$ are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, p \geq i} A_W^{p,q}, \quad \bar{F}^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, q \geq i} A_W^{p,q}.$$

Put

$$A_F^{p,q} := W_{p+q} A \cap F^p A \cap \left(\bar{F}^q A + \sum_{i>0} W_{p+q-i} A \cap \bar{F}^{q-i+1} A \right),$$

and, interchanging the roles of $F, \bar{F}$ and of p, q ,

$$A_{\bar{F}}^{p,q} := W_{p+q} A \cap \bar{F}^q A \cap \left(F^p A + \sum_{i>0} W_{p+q-i} A \cap F^{p-i+1} A \right).$$

For $p + q = n$, the projection from $W_n A$ to $\mathrm{Gr}_n^W A$ induces isomorphisms

$$A_F^{p,q} \simeq A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \simeq A_W^{p,q}.$$

It follows that the $A_F^{p,q}$, respectively the $A_{\bar{F}}^{p,q}$, form a bigrading of A . The sums of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

are bifiltered isomorphisms

$$a_F : (A; W, F) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, F_W), \quad a_{\bar{F}} : (A; W, \bar{F}) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, \bar{F}_W).$$

Lemma 1.1. *The automorphism $d = a_F a_{\overline{F}}^{-1}$ of $\mathrm{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r < p \\ s < q}} A_W^{r,s}. \quad (1.1.1)$$

Proof. Passing to the associated graded, the isomorphisms a_F and $a_{\overline{F}}$ induce the identity on $\mathrm{Gr}^W A$. Thus $\mathrm{Gr}^W(d) = 1$. For a subset $I \subset \mathbb{Z}^2$ and a bigraded object B , write $B_I = \bigoplus_{(r,s) \in I} B^{r,s}$. Let

$$I(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } r < p)\},$$

$$J(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } s < q)\},$$

$$K(p,q) = \{(r,s) : r+s \leq p+q \text{ and } r < p \text{ and } s < q\}.$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects, the first subobject is $A_{FI(p,q)}$ and the second is $A_{\overline{F}J(p,q)}$. Hence $dA_W^{p,q}$ lies in the intersection corresponding to $I(p,q) \cap J(p,q) = K(p,q) \cup \{(p,q)\}$. Since $\mathrm{Gr}^W(d) = 1$, this gives exactly (1.1.1). $\square$

Proposition 1.1. *The functor $A \mapsto \mathrm{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \overline{F}$ to the category of objects of $\mathcal{A}$ endowed with a finite bigrading and with an automorphism d satisfying (1.1.1).*

The isomorphism $a_F : A \rightarrow \mathrm{Gr}^W A$ respects W , sends F to F_W , and sends $\overline{F}$ to $d^{-1}\overline{F}_W$. Thus the functor Gr^W has the inverse

$$s : (B = \bigoplus B^{p,q}, d) \mapsto (B, W, F_W, d^{-1}\overline{F}_W),$$

with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \mathrm{Gr}^W A$. Conversely, starting from a bigraded object B with an automorphism d , define $W, F, \overline{F}$ as above. If d satisfies (1.1.1), the evident map $B \rightarrow \mathrm{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \mathrm{Gr}^W \circ s$.

Remark 1.1. *If 2 is invertible in the ground field, then d has a unique square root $d^{1/2}$ still satisfying (1.1.1). Put*

$$a := d^{1/2} a_F = d^{-1/2} a_{\overline{F}}.$$

Then $a_{\overline{F}} = d^{1/2} a$ and $a_F = d^{-1/2} a$; thus a sends F , respectively $\overline{F}$, to

$$d^{1/2}(F_W), \quad d^{-1/2}(\overline{F}_W)$$

on $\mathrm{Gr}^W A$. The isomorphism a is the same for $(A; W, F, \overline{F})$ and for $(A; W, \overline{F}, F)$.

1.4

The constructions are compatible with tensor products. If

$$\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{A}$$

is a biadditive exact functor, it is extended to filtered objects by

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a A_1 \otimes F^b A_2 \subset A_1 \otimes A_2.$$

The associated-graded functor is compatible with $\otimes$, and, for bifiltered objects, the canonical isomorphism

$$\mathrm{Gr}^W \mathrm{Gr}^F(A_1 \otimes A_2) \simeq \mathrm{Gr}^W \mathrm{Gr}^F(A_1) \otimes \mathrm{Gr}^W \mathrm{Gr}^F(A_2)$$

sends the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2 . For vector spaces this follows from the fact that every bifiltration underlies a bigrading. Hence, if $(W_i, F_i, \overline{F}_i)$ are three opposite filtrations on A_i , then $(W_1 \otimes W_2, F_1 \otimes F_2, \overline{F}_1 \otimes \overline{F}_2)$ are again opposite on $A_1 \otimes A_2$. The bigradings, the isomorphisms $a_F, a_{\overline{F}}$, and the automorphism d are all compatible with $\otimes$.

1.5

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k , and let $\mathcal{C}$ be the category of vector spaces endowed with three opposite filtrations $W, F, \overline{F}$. This is a rigid abelian tensor category with $\text{End}(1) = k$. It has the fiber functor ω_0 , the underlying vector space, hence it is neutral Tannakian. It also has the fiber functor $\text{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\overline{F}}$ are isomorphisms of fiber functors $\omega_0 \rightarrow \omega_W$. Put

$$\mathcal{G} = \text{Aut}(\omega_W).$$

The functorial automorphism d defines a k -point of the pro-algebraic group $\mathcal{G}$. The functorial bigrading of $\text{Gr}^W(V)$ by the $V^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow \mathcal{G}.$$

It is a section of the projection

$$\text{pr} : \mathcal{G} \longrightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion of the category of bigraded vector spaces in $\mathcal{C}$. We normalize α by

$$\alpha(\lambda, \mu)v = \lambda^p \mu^q v \quad (v \in V^{p,q}).$$

By Proposition 1.1, $\mathcal{G}$ is also the algebraic group of automorphisms of the fiber functor on the category of bigraded vector spaces endowed with an automorphism d satisfying (1.1.1). We now compute it when k has characteristic zero.

Let $\mathfrak{l}$ be the Lie algebra over k freely generated by elements

$$D^{i,j} \quad (i, j < 0),$$

with the unique bigrading in which $D^{i,j}$ has bidegree (i, j) . The $W_n \mathfrak{l}$ are ideals, and the quotients $\mathfrak{l}/W_n \mathfrak{l}$ are nilpotent Lie algebras. Let $\mathcal{U}$ be the pro-algebraic group which is the projective limit of the unipotent groups

$$\exp(\mathfrak{l}/W_n \mathfrak{l}).$$

Let $\mathbb{G}_m \times \mathbb{G}_m$ act on $\mathfrak{l}$ so that (λ, μ) acts on bidegree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects W and therefore gives an action of $\mathbb{G}_m \times \mathbb{G}_m$ on $\mathcal{U}$.

Construction 1.1. *The group $\mathcal{G}$ is the semidirect product*

$$\mathcal{G} = (\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U}.$$

Indeed, giving an action of this semidirect product on a finite-dimensional vector space V is equivalent to giving an action of $\mathbb{G}_m \times \mathbb{G}_m$, hence a bigrading, together with a compatible nilpotent action of $\mathfrak{l}$. Equivalently, it is the datum of endomorphisms $D^{i,j}$ of bidegree (i, j) , or of their sum

$$D = \sum D^{i,j},$$

with

$$D(V^{p,q}) \subset \bigoplus_{r < p, s < q} V^{r,s}.$$

Writing $d = \exp(D)$, this is precisely the datum of a bigrading and of an automorphism satisfying (1.1.1). The construction is compatible with tensor products, giving the required isomorphism

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U} \xrightarrow{\sim} \mathcal{G} = \text{Aut}(\omega_W).$$

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space endowed with an increasing filtration W , whose complexification $V_{\mathbb{C}}$ is endowed with a decreasing filtration F , such that the filtration $W_{\mathbb{C}}$, the filtration F , and its complex conjugate $\bar{F}$ form a system of three opposite filtrations.

Via the dictionary between real vector spaces and complex vector spaces with a real structure, giving a real mixed Hodge structure is equivalently giving a complex vector space with three opposite filtrations $W, F, \bar{F}$ and with a complex conjugation respecting W and interchanging F and $\bar{F}$. By Proposition 1.1, the functor Gr^W identifies this category with the category of bigraded complex vector spaces

$$V = \bigoplus V^{p,q}$$

equipped with an automorphism d satisfying (1.1.1) and a complex conjugation σ interchanging $V^{p,q}$ and $V^{q,p}$, for which the complex conjugate of d is d^{-1} :

$$\sigma d \sigma^{-1} = d^{-1}.$$

Let $\mathcal{H}$ be the category of real mixed Hodge structures. It is a rigid abelian tensor category with $\text{End}(1) = \mathbb{R}$. It has the fiber functor ω_0 , the underlying real vector space, hence is neutral Tannakian. It also has the fiber functor $\text{Gr}^W =: \omega_W$. With the notation of Remark 1.1, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \text{Gr}^W(V_{\mathbb{C}})$$

respects the real structures on both sides, since it depends symmetrically on F and $\bar{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, put $k = \mathbb{C}$, and let $\mathcal{G}_{\mathbb{R}}$ be the real form of $\mathcal{G}$ defined by the following complex conjugation:

- (a) on $\mathbb{G}_m \times \mathbb{G}_m$, it is $(\lambda, \mu) \mapsto (\bar{\mu}, \bar{\lambda})$;
- (b) on $\mathcal{U}$, it is the one compatible with the complex conjugation of $\mathfrak{l}$ given by

$$D^{i,j} \longmapsto -D^{j,i}.$$

As in §1.1, giving an action of $\mathcal{G}_{\mathbb{R}}$ on a real vector space V is equivalent to giving:

- (a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another;
- (b) an automorphism d of $V_{\mathbb{C}}$ such that $\bar{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{i < p, j < q} V^{i,j}.$$

Proposition 2.1. *The fiber functor ω_W induces an equivalence between $\mathcal{H}$ and the category of representations of $\mathcal{G}_{\mathbb{R}}$. Thus*

$$\mathcal{G}_{\mathbb{R}} = \text{Aut}(\omega_W).$$

The inverse functor is

$$(V, \text{ action of } \mathcal{G}_{\mathbb{R}}) \longmapsto (V, \text{ bigrading of } V_{\mathbb{C}}, d) \longmapsto (V, W, F),$$

where W is the real filtration whose complexification is the weight filtration attached to the bigrading of $V_{\mathbb{C}}$, and

$$F = d^{-1/2}(F_W)$$

on the complexification. The isomorphism a gives the functorial identification

$$\phi \circ \text{Gr}^W \xrightarrow{\sim} \text{Id}.$$